

Topic 2

A classification problem is formed by two classes. We are given a set of 2-dimensional data $\mathbf{X} = [\mathbf{x}_1 \mathbf{x}_2 \mathbf{x}_3 \mathbf{x}_4]$:

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & -1 & -1 \end{bmatrix} \quad (1)$$

each belonging to one of the two classes, as indicated in the class label vector $\mathbf{l} = [l_1 \ l_2 \ l_3 \ l_4]$:

$$\mathbf{l} = [1 \ 1 \ 2 \ 2]. \quad (2)$$

Using the above (training) vectors and the corresponding class labels, classify the following vectors:

$$\mathbf{x}_5 = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad \mathbf{x}_6 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{x}_7 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \mathbf{x}_8 = \begin{bmatrix} -2 \\ 0 \end{bmatrix} \quad (3)$$

using the following classifiers:

- 1) (Score: 25%) The Nearest Class Centroid (NCC) classifier.
- 2) (Score: 25%) The Nearest Neighbor Classifier using only one neighbor.

Classification Question 1

First I find the centroid of each class, i.e. the mean of the training data in that class:

$$\mu_1 = ([1, 2]^T + [1, 0]^T)/2 = [1, 1]^T$$

$$\mu_2 = ([0, -1]^T + [-2, -1]^T)/2 = [-1, -1]^T$$

Then to classify each point I pick the class whose centroid is nearest to the new data point:

$$\|\mu_1 - x_5\|_2 = \sqrt{2}$$

$$\|\mu_2 - x_5\|_2 = \sqrt{10}$$

$$\therefore x_5 \in C_1$$

$$\|\mu_1 - x_6\|_2 = \sqrt{2}$$

$$\|\mu_2 - x_6\|_2 = \sqrt{2}$$

For x_6 either class could be chosen

$$\|\mu_1 - x_7\|_2 = 2$$

$$\|\mu_2 - x_7\|_2 = 2$$

For x_7 either class could be chosen

$$\|\mu_1 - x_8\|_2 = \sqrt{10}$$

$$\|\mu_2 - x_8\|_2 = \sqrt{2}$$

$$\therefore x_8 \in C_2$$

Classification Question 2

For every new datapoint, the euclidean distance to each point in the training data is calculated, the class corresponding to the training point that is closest to the new datapoint is chosen.

For x_5 the distances to each datapoint are:

$$\|x_1 - x_5\|_2 = \sqrt{5}$$

$$\|x_2 - x_5\|_2 = 1 \leftarrow \text{smallest}$$

$$\|x_3 - x_5\|_2 = \sqrt{5}$$

$$\|x_4 - x_5\|_2 = \sqrt{17}$$

$$\therefore x_5 \in C_1$$

For x_6 the distances to each datapoint are:

$$\|x_1 - x_6\|_2 = \sqrt{2}$$

$$\|x_2 - x_6\|_2 = 1$$

$$\|x_3 - x_6\|_2 = 1$$

$$\|x_4 - x_6\|_2 = \sqrt{5}$$

$\therefore x_6$ could be in either class

For x_7 the distances to each datapoint are:

$$\|x_1 - x_7\|_2 = \sqrt{5}$$

$$\|x_2 - x_7\|_2 = \sqrt{5}$$

$$\|x_3 - x_7\|_2 = \sqrt{5}$$

$$\|x_4 - x_7\|_2 = \sqrt{5}$$

$\therefore x_7$ could be in either class

For x_8 the distances to each datapoint are:

$$\|x_1 - x_8\|_2 = \sqrt{13}$$

$$\|x_2 - x_8\|_2 = \sqrt{9}$$

$$\|x_3 - x_8\|_2 = \sqrt{5}$$

$$\|x_4 - x_8\|_2 = 1 \leftarrow \text{smallest}$$

$$\therefore x_8 \in C_2$$

Note that the two classifiers agree in this case.

LDA Question 1

1) (Score: 25%) Calculate the projection vector $w \in \mathbb{R}^2$ obtained by the training data.

$\mu_1 = [1, 1]^T, \mu_2 = [-1, -1]^T$ from the previous questions

$$\mu = (\mu_1 + \mu_2)/2 = [0, 0]^T$$

Single-class scatter matrices:

$$S_1 = (\mathbf{x}_1 - \mu_1)(\mathbf{x}_1 - \mu_1)^T + (\mathbf{x}_2 - \mu_1)(\mathbf{x}_2 - \mu_1)^T$$

$$S_1 = [0, 1]^T[0, 1] + [0, -1]^T[0, -1]$$

$$S_1 = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

$$S_2 = [1, 0]^T[1, 0] + [-1, 0]^T[-1, 0]$$

$$S_2 = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

Within class scatter matrix:

$$S_W = S_1 + S_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Between class scatter matrix:

$$S_B = 2(\mu_1 - \mu)(\mu_1 - \mu)^T + 2(\mu_2 - \mu)(\mu_2 - \mu)^T$$

$$S_B = 2[1, 1]^T[1, 1] + 2[-1, -1]^T[-1, -1]$$

$$S_B = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$$

Combining them:

$$S_W^{-1}S_B = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

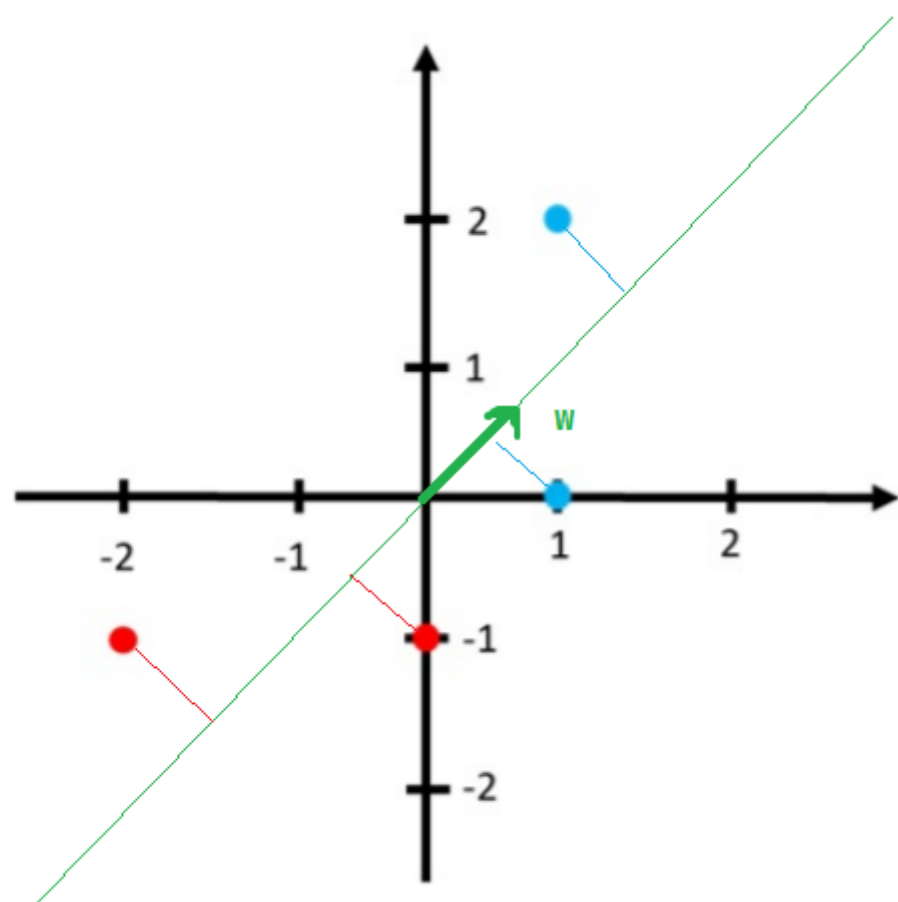
Next, finding the biggest eigenvalue and associated eigenvector:

Using a calculator, I've determined the largest eigenvalue to be $\lambda_1 = 4$, and the associated eigenvector $v_1 = [1, 1]^T$, however we want a unit vector for projection vector, so I scale it:

$$w = \frac{v_1}{||v_1||} = [\sqrt{2}/2, \sqrt{2}/2]^T$$

LDA Question 2

2) (Score: 25%) Draw in Figure 2 the projection vector of LDA w calculated in question 1.



Week 7) Book exercise 4.7)

By definition, it holds that

$$\begin{aligned}\sigma(a) + \sigma(-a) &\stackrel{\text{def.}}{=} \frac{1}{1+\exp(a)} + \frac{1}{1+\exp(-a)} \\ &= \frac{2 + \exp(a) + \exp(-a)}{(1+\exp(-a))(1+\exp(a))} \\ &= \frac{2 + \exp(a) + \exp(-a)}{2 + \exp(a) + \exp(-a)} = 1\end{aligned}$$

$$\Rightarrow \sigma(-a) = 1 - \sigma(a).$$

Let $y = \sigma(a)$. For the inverse, we have

$$a = \frac{1}{1 + \exp(-y)}$$

$$\Rightarrow \exp(-y) = \frac{1-a}{a}$$

$$\Rightarrow y = \ln\left(\frac{a}{1-a}\right).$$

Book exercise 4.9)

Note that we have a likelihood function

$$\begin{aligned} p(\{q_n, t_{n,r}\} | \Pi_1, \dots, \Pi_K) &= \prod_{n=1}^N \prod_{r=1}^K [p(q_n | e_n) p(e_n)]^{t_{n,r}} \\ &= \prod_{n=1}^N \prod_{r=1}^K [\Pi_r p(q_n | e_n)]^{t_{n,r}} \end{aligned}$$

and thus a log-likelihood

$$\begin{aligned} \ln(p) &= \sum_{n=1}^N \sum_{r=1}^K t_{n,r} [\ln(\Pi_r) + \ln(p(q_n | e_n))] \\ &\propto \sum_{n=1}^N \sum_{r=1}^K t_{n,r} \ln(\Pi_r) \end{aligned}$$

"proportional to"

is enough as our
interest is in optimizing Π_r

Note that Π_r represents a probability.
Thus, it is constrained $\Pi_r \leq 1$, $\sum_{r=1}^K \Pi_r = 1$, and we
invoke the Lagrangian

$$2 = \left(\sum_{n=1}^N \sum_{r=1}^K t_{n,r} \ln(\Pi_r) \right) + \lambda \cdot \left(\left(\sum_{r=1}^K \Pi_r \right) - 1 \right)$$

Taking the derivative and setting equal
to zero yields

$$\frac{\partial 2}{\partial \Pi_r} = \sum_{n=1}^N \frac{t_{n,r}}{\Pi_r} + \lambda = 0$$

$$\Rightarrow \pi_k = - \frac{\sum_{n=1}^N b_{n,k}}{\lambda}$$

$$= - \frac{M_k}{\lambda} \quad (*)$$

By def.

Now, summing over k gives

$$\sum_{k=1}^N \pi_k = \sum_{k=1}^N \left(- \frac{M_k}{\lambda} \right)$$

$$\Rightarrow 1 = \frac{N}{-\lambda}$$

$$\Rightarrow \lambda = -N.$$

Which in (*) again yields

$$\pi_k = \frac{M_k}{N}$$

(note also here we "for free" have that $\pi_k \geq 0$).

Book exercise 4.12)

Using the chain rule, we have that

$$\frac{d\sigma(a)}{da} = \frac{\exp(a)}{(1+\exp(a))^2}$$

$$= \frac{\exp(a)}{1+\exp(a)} \cdot \frac{1}{1+\exp(a)}$$

$$= (1-\sigma(a)) \sigma(a)$$

Book exercise 4.13)

By (4.88), it follows that

$$\begin{aligned}\nabla E(\underline{w}) &= -\nabla \left\{ \sum_{n=1}^N (t_n \ln(y_n) + (1-t_n) \ln(1-y_n)) \right\} \\ &= -\sum_{n=1}^N \nabla \{ t_n \ln(y_n) + (1-t_n) \ln(1-y_n) \} \\ &= -\sum_{n=1}^N \left\{ \frac{\partial}{\partial y_n} (t_n \ln(y_n) + (1-t_n) \ln(1-y_n)) \right. \\ &\quad \left. \cdot \frac{\partial y_n}{\partial a_n} \cdot \frac{\partial a_n}{\partial \underline{w}} \right\}\end{aligned}$$

Multiple applications
of chain rule

Using (4.88)

$$\begin{aligned}&= -\sum_{n=1}^N \left(\frac{t_n}{y_n} - \frac{1-t_n}{1-y_n} \right) \cdot y_n \cdot (1-y_n) \underline{\phi}_n \\ &= -\sum_{n=1}^N \frac{t_n - y_n}{y_n(1-y_n)} y_n (1-y_n) \underline{\phi}_n \\ &= \sum_{n=1}^N (y_n - t_n) \underline{\phi}_n.\end{aligned}$$

Linear classification (out-of-book exercise)

As usual, we omit suggested solutions for nearest class centroid and nearest neighbor classifiers.

For the Fisher's discriminant analysis part, it is simply a matter of reading section 4.1.4 and using equations (4.2.1) through (4.30).